

Cada componente incluye

HTML
plantilla,
TS
clase y
SCSS
estilos.

Instalación

CLI

```
ng add @angular/material
// instala CDK + tema + fuente de iconos
```

TS

app.config.ts (Angular 22+)

```
// Material 22 ya NO requiere provider de
// animaciones: anima con CSS por defecto.
// (provideAnimationsAsync esta deprecado
// desde v20.2, se elimina en v23)
export const appConfig: ApplicationConfig = {
  providers: [ /* provideHttpClient(), ... */ ]
};
```

SCSS

styles.scss (tema M3)

```
@use '@angular/material' as mat;
html {
  @include mat.theme({
    color: mat.$rose-palette,
    typography: Roboto,
    density: 0,
  });
}
```

Montaje standalone (base)

El patrón: cada componente importa sus MatXxxModule en imports:[].

TS

```
import { Component } from '@angular/core';
import { MatButtonModule }
  from '@angular/material/button';
import { MatCardModule }
  from '@angular/material/card';

@Component({
  selector: 'app-demo',
  standalone: true,
  imports: [MatButtonModule, MatCardModule],
  templateUrl: './demo.component.html',
  styleUrls: ['./demo.component.scss']
})
export class DemoComponent {}
```

HTML

```
<mat-card>
  <button mat-raised-button
    color="primary">Hola</button>
</mat-card>
```

Button

import {MatButtonModule} from '@angular/material/button'

HTML

```
<button mat-button>Basic</button>
<button mat-raised-button color="primary"
  (click)="save()">Raised</button>
<button mat-flat-button>Flat</button>
<button mat-stroked-button>Stroked</button>
<button mat-icon-button>
  <mat-icon>favorite</mat-icon></button>
<button mat-fab><mat-icon>add</mat-icon></button>
```

TS

```
export class DemoComponent {
  save(): void { console.log('guardado'); }
}
```

SCSS

```
button { margin: 4px; }
.mat-mdc-fab { --mdc-fab-container-color: #DD0031; }
```

Icon

import {MatIconModule} from '@angular/material/icon'

HTML

```
<mat-icon>home</mat-icon>
<mat-icon fontIcon="settings"></mat-icon>
<mat-icon svgIcon="logo"></mat-icon>
```

TS

(registrar SVG propio)

```
constructor(
  private reg: MatIconRegistry,
  private san: DomSanitizer) {
  this.reg.addSvgIcon('logo',
    this.san.bypassSecurityTrustResourceUrl(
      'assets/logo.svg'));
}
```

SCSS

```
mat-icon { color: #DD0031; }
mat-icon.big { font-size: 48px;
  width: 48px; height: 48px; }
```

Card

import {MatCardModule} from '@angular/material/card'

HTML

```
<mat-card>
  <mat-card-header>
    <mat-card-title>Titulo</mat-card-title>
    <mat-card-subtitle>Sub</mat-card-subtitle>
  </mat-card-header>
  <mat-card-content>Contenido</mat-card-content>
  <mat-card-actions>
    <button mat-button>OK</button>
  </mat-card-actions>
</mat-card>
```

TS

```
export class DemoComponent {
  title = 'Titulo';
}
```

SCSS

```
mat-card { max-width: 400px; margin: 12px; }
mat-card-actions { justify-content: flex-end; }
```

Form Field + Input

import {MatFormFieldModule} de ../form-field
import {MatInputModule} de ../input

HTML

```
<mat-form-field appearance="outline">
  <mat-label>Nombre</mat-label>
  <input matInput [formControl]="name">
  <mat-hint>Ayuda</mat-hint>
  <mat-error *ngIf="name.hasError('required')">
    Requerido</mat-error>
</mat-form-field>
```

TS

```
import { FormControl, Validators }
  from '@angular/forms';
export class DemoComponent {
  name = new FormControl('',
    [Validators.required]);
}
```

SCSS

```
mat-form-field { width: 100%; }
```

Select

```
import {MatSelectModule} from '@angular/material/select'
```

HTML

```
<mat-form-field>
  <mat-label>Sede</mat-label>
  <mat-select [(value)]="sede">
    <mat-option *ngFor="let s of sedes"
      [value]="s.id">{{s.nombre}}</mat-option>
  </mat-select>
</mat-form-field>
```

TS

```
export class DemoComponent {
  sede = 'Lima';
  sedes = [
    { id: 'Lima', nombre: 'Lima' },
    { id: 'cusco', nombre: 'Cusco' },
  ];
}
```

SCSS

```
mat-form-field { min-width: 200px; }
```

Autocomplete

```
import {MatAutocompleteModule} de ../autocomplete
```

HTML

```
<mat-form-field>
  <input matInput [formControl]="ctrl"
    [matAutocomplete]="a">
</mat-form-field>
<mat-autocomplete #a="matAutocomplete">
  <mat-option *ngFor="let o of filtered" async
    [value]="o">{{o}}</mat-option>
</mat-autocomplete>
```

TS

```
ctrl = new FormControl('');
opts = ['Uno','Dos','Tres'];
filtered = this.ctrl.valueChanges.pipe(
  startWith(''),
  map(v => this.opts.filter(o =>
    o.toLowerCase().includes(
      (v||'').toLowerCase()))));
```

SCSS

```
.mat-mdc-autocomplete-panel { max-height: 240px; }
```

Checkbox / Radio / Toggle

HTML

```
<mat-checkbox [(ngModel)]="ok">Acepto</mat-checkbox>
<mat-radio-group [(ngModel)]="g">
  <mat-radio-button value="m">M</mat-radio-button>
  <mat-radio-button value="f">F</mat-radio-button>
</mat-radio-group>
<mat-slide-toggle [(ngModel)]="on">
  Activar</mat-slide-toggle>
```

TS

```
// imports: MatCheckboxModule, MatRadioModule,
//           MatSlideToggleModule, FormsModule
ok = false;
g = 'm';
on = true;
```

SCSS

```
mat-radio-button { margin-right: 12px; }
```

Slider

```
import {MatSliderModule} from '@angular/material/slider'
```

HTML

```
<mat-slider min="0" max="100" step="1">
  <input matSliderThumb [(ngModel)]="vol">
</mat-slider>
```

TS

```
export class DemoComponent { vol = 50; }
```

SCSS

```
mat-slider { width: 100%; }
```

Datepicker

```
MatDatepickerModule + MatNativeDateModule (de ../core)
```

HTML

```
<mat-form-field>
  <mat-label>Fecha</mat-label>
  <input matInput [matDatepicker]="p"
    [(ngModel)]="fecha">
  <mat-datepicker-toggle matIconSuffix
    [for]="p"></mat-datepicker-toggle>
  <mat-datepicker #p></mat-datepicker>
</mat-form-field>
```

TS

```
import { provideNativeDateAdapter }
  from '@angular/material/core';
@Component({
  providers: [ provideNativeDateAdapter() ],
})
export class DemoComponent { fecha = new Date(); }
```

SCSS

```
mat-form-field { width: 220px; }
```

Chips

```
import {MatChipsModule} from '@angular/material/chips'
```

HTML

```
<mat-chip-set>
  <mat-chip *ngFor="let t of tags"
    (removed)="remove(t)">
    {{t}}
  <button matChipRemove>
    <mat-icon>cancel</mat-icon></button>
</mat-chip>
</mat-chip-set>
```

TS

```
tags = ['Angular','Material'];
remove(t: string) {
  this.tags = this.tags.filter(x => x !== t);
}
```

SCSS

```
mat-chip { margin: 2px; }
```

List

```
import {MatListModule} from '@angular/material/list'
```

HTML

```
<mat-list>
  <mat-list-item *ngFor="let i of items">
    {{i}}</mat-list-item>
</mat-list>
<mat-nav-list>
  <a mat-list-item routerLink="/home">Home</a>
</mat-nav-list>
```

TS

```
items = ['Item_1','Item_2','Item_3'];
```

SCSS

```
mat-list { border: 1px solid #eee;
  border-radius: 8px; }
```

Toolbar

```
import {MatToolbarModule} from '@angular/material/toolbar'
```

HTML

```
<mat-toolbar color="primary">
  <span>Mi App</span>
  <span class="spacer"></span>
  <button mat-icon-button (click)="menu()">
    <mat-icon>menu</mat-icon></button>
</mat-toolbar>
```

TS

```
menu(): void { /* abrir sidenav */ }
```

SCSS

(el truco del spacer)

```
.spacer { flex: 1 1 auto; }
```

Sidenav

import {MatSidenavModule} from '@angular/material/sidenav'

HTML

```
<mat-sidenav-container>
  <mat-sidenav #s mode="side" opened>
    Menu lateral</mat-sidenav>
  <mat-sidenav-content>
    <button (click)="s.toggle()">Toggle</button>
  </mat-sidenav-content>
</mat-sidenav-container>
```

TS

```
// #s es una referencia de plantilla,
// no requiere logica en el .ts para toggle()
```

SCSS

```
mat-sidenav-container { height: 100vh; }
mat-sidenav { width: 240px; }
```

Tabs

import {MatTabsModule} from '@angular/material/tabs'

HTML

```
<mat-tab-group [(selectedIndex)]="tab">
  <mat-tab label="Uno">Contenido 1</mat-tab>
  <mat-tab label="Dos">Contenido 2</mat-tab>
</mat-tab-group>
```

TS

```
export class DemoComponent { tab = 0; }
```

SCSS

```
.mat-mdc-tab-body-content { padding: 16px; }
```

Menu

import {MatMenuModule} from '@angular/material/menu'

HTML

```
<button mat-button [matMenuTriggerFor]="m">
  Opciones</button>
<mat-menu #m="matMenu">
  <button mat-menu-item (click)="edit()">
    Editar</button>
  <button mat-menu-item (click)="del()">
    Borrar</button>
</mat-menu>
```

TS

```
edit(): void { /* ... */ }
del(): void { /* ... */ }
```

SCSS

```
.mat-mdc-menu-item mat-icon { margin-right: 8px; }
```

Expansion Panel

import {MatExpansionModule} de ../expansion

HTML

```
<mat-accordion>
  <mat-expansion-panel [(expanded)]="open">
    <mat-expansion-panel-header>
      <mat-panel-title>Seccion</mat-panel-title>
    </mat-expansion-panel-header>
    <p>Detalle</p>
  </mat-expansion-panel>
</mat-accordion>
```

TS

```
export class DemoComponent { open = true; }
```

SCSS

```
mat-expansion-panel { margin-bottom: 8px; }
```

Progress

MatProgressBarModule · MatProgressSpinnerModule

HTML

```
<mat-progress-bar mode="determinate"
  [value]="pct"></mat-progress-bar>
<mat-spinner *ngIf="loading"
  diameter="30"></mat-spinner>
```

TS

```
pct = 40;
loading = false;
```

SCSS

```
mat-progress-bar { --mdc-linear-progress-active-indicator-color: #DD0031; }
```

Table

import {MatTableModule} from '@angular/material/table'

HTML

```
<table mat-table [dataSource]="data">
  <ng-container matColumnDef="id">
    <th mat-header-cell *matHeaderCellDef>ID</th>
    <td mat-cell *matCellDef="let r">{{r.id}}</td>
  </ng-container>
  <tr mat-header-row
    *matHeaderRowDef="cols"></tr>
  <tr mat-row
    *matRowDef="let r, columns: columns"></tr>
</table>
```

TS

```
cols = ['id', 'nombre'];
data = [
  { id: 1, nombre: 'Ana' },
  { id: 2, nombre: 'Luis' },
];
```

SCSS

```
table { width: 100%; }
.mat-mdc-header-cell { font-weight: 700; }
```

Paginator

import {MatPaginatorModule} de ../paginator

HTML

```
<mat-paginator [length]="100"
  [pageSize]="10"
  [pageSizeOptions]="[5,10,25]"
  (page)="onPage($event)">
</mat-paginator>
```

TS

```
onPage(e: PageEvent) {
  console.log(e.pageIndex, e.pageSize);
}
```

SCSS

```
mat-paginator { background: transparent; }
```

Dialog

import {MatDialogModule, MatDialog} de ../dialog

TS

(componente que abre)

```
constructor(private dlg: MatDialog) {}
open() {
  const ref = this.dlg.open(MiDialogComp,
    { width: '400px', data: { id: 1 } });
  ref.afterClosed().subscribe(r =>
    console.log('cerrado', r));
}
```

HTML

(plantilla del dialog)

```
<h2 mat-dialog-title>Titulo</h2>
<mat-dialog-content>...</mat-dialog-content>
<mat-dialog-actions>
  <button mat-button mat-dialog-close>
    Cerrar</button>
</mat-dialog-actions>
```

SCSS

```
.mat-mdc-dialog-content { min-height: 80px; }
```

Snackbar

```
import {MatSnackBarModule, MatSnackBar} de .../snack-bar
```

TS

```
constructor(private sb: MatSnackBar) {}  
notify() {  
  this.sb.open('Guardado!', 'OK',  
    { duration: 3000,  
      horizontalPosition: 'end' });  
}
```

SCSS

```
.mat-mdc-snack-bar-container {  
  --mdc-snack-bar-container-color: #1A1A1A; }
```

Tooltip / Badge / Divider

HTML

```
<button matTooltip="Info_util">Hover</button>  
<button matBadge="4" matBadgeColor="warn">  
  <mat-icon>notifications</mat-icon></button>  
<mat-divider></mat-divider>
```

TS

```
// imports: MatTooltipModule, MatBadgeModule,  
//          MatDividerModule  
count = 4;
```

SCSS

```
mat-divider { margin: 12px 0; }
```

Imports más usados

Componente	Módulo
Botón	MatButtonModule
Ícono	MatIconModule
Card	MatCardModule
Input	MatInputModule
Select	MatSelectModule
Tabla	MatTableModule
Dialog	MatDialogModule
Snackbar	MatSnackBarModule
Toolbar	MatToolbarModule

Buenas prácticas

- ✓ Un MatXxxModule por componente que uses (tree-shaking).
- ✓ Usá appearance="outline" en form-fields para look M3.
- ✓ Definí el tema en styles.scss con mat.theme.
- ! DatePicker necesita provideNativeDateAdapter().
- ! Para overridear estilos MDC usá las clases .mat-mdc-* y variables CSS.